

Assignment – 1 (TCP)

// A Java program for a Client

```
import java.net.*;
import java.io.*;

public class Client
{
    private Socket socket = null;
    private DataInputStream input = null;
    private DataOutputStream out = null;

    public Client(String address, int port)
    {
        try
        {
            socket = new Socket(address, port);
            System.out.println("Connected");

            input = new DataInputStream(System.in);
            out = new DataOutputStream(socket.getOutputStream());
        }
        catch(UnknownHostException u)
        {
            System.out.println(u);
        }
        catch(IOException i)
        {
            System.out.println(i);
        }

        String line = "";
        while (!line.equals("Over"))
        {
            try
            {
                line = input.readLine();
                out.writeUTF(line);
            }
            catch(IOException i)
            {
                System.out.println(i);
            }
        }

        try {
            input.close();
            out.close();
            socket.close();
        }
        catch(IOException i)
        {
            System.out.println(i);
        }
    }

    public static void main(String args[])
    {
        Client client = new Client("localhost", 8080);
    }
}
```

```
{ Client client = new Client("127.0.0.1", 5000); }
```

// A Java program for a Server

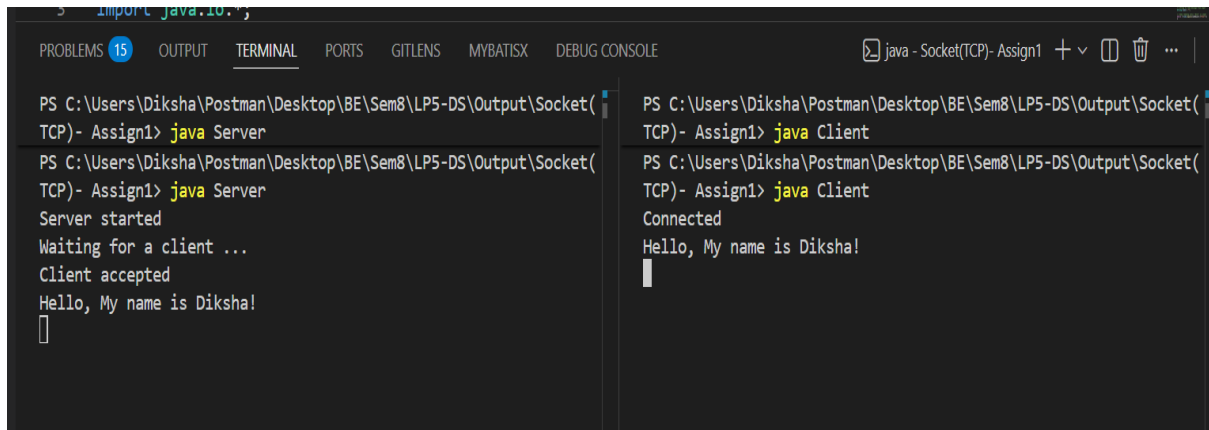
```
import java.net.*;
import java.io.*;

public class Server
{
    private Socket    socket = null;
    private ServerSocket server = null;
    private DataInputStream in  = null;

    public Server(int port)
    {
        try    {server = new ServerSocket(port);
                System.out.println("Server started");
                System.out.println("Waiting for a client ...");
                socket = server.accept();
                System.out.println("Client accepted");
                in = new DataInputStream(
                    new BufferedInputStream(socket.getInputStream()));
                String line = "";
                while (!line.equals("Over"))
                {
                    try
                    {
                        line = in.readUTF();
                        System.out.println(line);
                    }
                    catch(IOException i)
                    {
                        System.out.println(i);
                    }
                }
                System.out.println("Closing connection");
                socket.close();
                in.close();
            }
            catch(IOException i)
            {
                System.out.println(i);
            }
        }

    public static void main(String args[])
    {
        Server server = new Server(5000);
    }
}
```

Output:-



```
PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output\Socket(TCP)- Assign1> java Server
Server started
Waiting for a client ...
Client accepted
Hello, My name is Diksha!

PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output\Socket(TCP)- Assign1> java Client
Connected
Hello, My name is Diksha!
```

2) UDP :-

```
import java.net.*;
```

```
class QuoteClient {
```

```
    public static void main(String args[]) throws Exception {
```

```
        if (args.length < 2) {
```

```
            System.out.println("Usage: java QuoteClient <Host> <Port>");
```

```
            return; }

```

```
    InetAddress address = InetAddress.getByName(args[0]);
```

```
    int port = Integer.parseInt(args[1]);
```

```
    DatagramSocket socket = new DatagramSocket();
```

```
    byte[] send = "Give me quote".getBytes();
```

```
    byte[] receive = new byte[1024];
```

```
    while (true) {
```

```
        DatagramPacket request =
```

```
            new DatagramPacket(send, send.length, address, port);
```

```
        socket.send(request);
```

```
        DatagramPacket response =
```

```
            new DatagramPacket(receive, receive.length);
```

```
        socket.receive(response);
```

```
        String quote = new String(response.getData(), 0, response.getLength());
```

```
        System.out.println("Quote from Server: " + quote);
```

```
        Thread.sleep(2000); // wait 2 seconds    } }

```

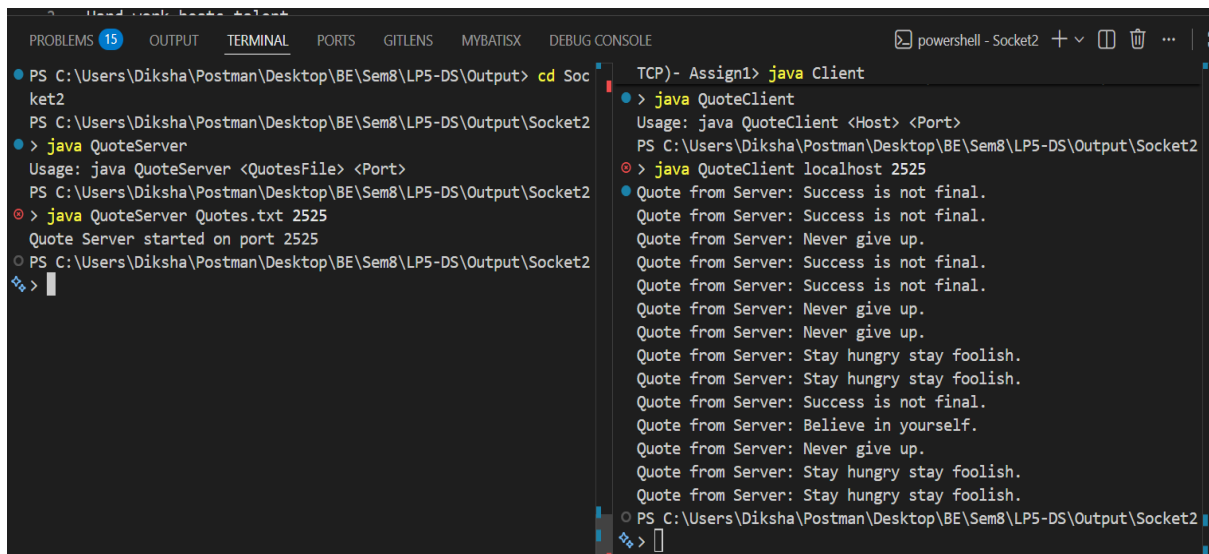
```
import java.io.*;
```

```

import java.net.*;
import java.util.*;
class QuoteServer {
    public static void main(String args[]) throws Exception {
        if (args.length < 2) {
            System.out.println("Usage: java QuoteServer <QuotesFile> <Port>");
            return;    }
        int port = Integer.parseInt(args[1]);
        DatagramSocket socket = new DatagramSocket(port);
        List<String> quotes = new ArrayList<>();
        BufferedReader br = new BufferedReader(new FileReader(args[0]));
        String line;
        while ((line = br.readLine()) != null) {
            quotes.add(line);    }
        br.close();
        System.out.println("Quote Server started on port " + port);
        byte[] receive = new byte[1024];
        Random random = new Random();
        while (true) {
            DatagramPacket request = new DatagramPacket(receive, receive.length);
            socket.receive(request);
            String quote = quotes.get(random.nextInt(quotes.size()));
            byte[] send = quote.getBytes();
            DatagramPacket response = new DatagramPacket(
                send,
                send.length,
                request.getAddress(),
                request.getPort());
            socket.send(response);    } }

```

Output:-



```
PROBLEMS 15 OUTPUT TERMINAL PORTS GITLENS MYBATISX DEBUG CONSOLE powershell - Socket2 + v [] [] ... |
● PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output> cd Socket2
PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output\Socket2>
● > java QuoteServer
Usage: java QuoteServer <QuotesFile> <Port>
PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output\Socket2>
● > java QuoteServer Quotes.txt 2525
Quote Server started on port 2525
● PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output\Socket2>
❖ >

TCP)- Assign1> java Client
● > java QuoteClient
Usage: java QuoteClient <Host> <Port>
PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output\Socket2>
● > java QuoteClient localhost 2525
● Quote from Server: Success is not final.
Quote from Server: Success is not final.
Quote from Server: Never give up.
Quote from Server: Success is not final.
Quote from Server: Success is not final.
Quote from Server: Never give up.
Quote from Server: Never give up.
Quote from Server: Stay hungry stay foolish.
Quote from Server: Stay hungry stay foolish.
Quote from Server: Success is not final.
Quote from Server: Believe in yourself.
Quote from Server: Never give up.
Quote from Server: Stay hungry stay foolish.
Quote from Server: Stay hungry stay foolish.
● PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output\Socket2>
❖ >
```